

## Title shortlist

1. **MedAgent: An LLM-Orchestrated System Integrating Multi-Disease ML Predictions with Multi-Method Explainability for Clinical Reporting**
2. **Agentic LLM Orchestration of Multi-Disease Diagnostic Models: A Unified Framework for Explainable, Confidence-Aware Medical Reporting**
3. **A Multi-Modal LLM-Agent Framework for Orchestrating Disease-Specific ML Models and Generating Explainable Medical Reports**
4. **From Symptoms to Summary: A Multi-Modal AI Agent for Integrating Text, Image, and Audio Inputs Across Nine Disease Prediction Modules**
5. **Towards Smart Healthcare Infrastructure: An LLM-Agent Pipeline for Multi-Modal Symptom Analysis and Multi-Disease Risk Prediction**

Top pick: #1 or #2 — "agent," "explainability," and "confidence-aware" are the keywords that signal your real contribution to reviewers.

## Full paper structure

Title, Authors, Abstract

Abstract (200-250 words): gap → your agentic architecture → 9 disease modules, multi-modal input → multi-method explainability (SHAP+LIME+more) fused into LLM reporting → key results → novelty statement in one explicit sentence. No table/figure.

### 1. Introduction (~1 page)

- Real-world gap: fragmented single-disease tools, raw unexplained probabilities, no multi-modal symptom integration.
- Contributions as 5-6 bullets: multi-modal ingestion; 9-module ML backbone; multi-method explainability fusion (SHAP, LIME, counterfactuals, uncertainty quantification); LLM-as-agent orchestration with confidence-aware routing; guideline-grounded, hallucination-checked reporting; end-to-end evaluation of both prediction and report quality.
- **Figure 1:** high-level system teaser (redrawn from your sketch).

### 2. Related Work (~1-1.5 pages)

- 2.1 ML/DL per disease category (metabolic, cardiac, imaging, dermatological, ophthalmic).
- 2.2 LLMs in healthcare / clinical report generation / medical agents.
- 2.3 Explainable AI in medicine (SHAP, LIME, Grad-CAM, counterfactuals).
- 2.4 Multi-agent / tool-using LLM architectures.

- **Table 1: Comparative summary of related systems** — System, Diseases Covered, Modalities, XAI Methods, LLM Integration, Multi-Agent?, Guideline Grounding?, Limitations (8-12 rows). This table visually proves your gap.

### 3. Proposed System Architecture (~2-2.5 pages — core section)

- 3.1 Overview — full walkthrough of the pipeline.
- 3.2 Multi-modal input layer — text/symptom parsing (NER for symptoms/duration/severity), ASR for audio (with confidence scores passed forward), image preprocessing.
- 3.3 Disease-specific ML modules (M1-M9) — algorithm, features, output per module.
- 3.4 **Explainability layer (multi-method)**:
  - 3.4.1 Tabular modules (diabetes, heart, liver, kidney): SHAP, LIME, PDP/ICE, counterfactuals (DiCE), Anchor explanations, SHAP interaction values.
  - 3.4.2 Image modules (X-ray, MRI, skin, eye): Grad-CAM/Grad-CAM++, LIME-for-images, Integrated Gradients, DeepSHAP/GradientSHAP, lesion localization.
  - 3.4.3 **Evidence fusion for LLM grounding** — how SHAP + LIME + counterfactuals + similar-case retrieval + guideline snippets are merged into one structured JSON per module before reaching the LLM. This is your strongest novelty claim.
- 3.5 Uncertainty quantification — Monte Carlo dropout / conformal prediction / ensemble variance, replacing raw softmax confidence.
- 3.6 LLM-as-Agent orchestration — routing logic, multi-module aggregation, cross-module correlation detection (e.g., diabetes + kidney → nephropathy link), confidence-weighted decision on definitive vs. cautious language.
- 3.7 Report generation — prompt template, guideline-grounded phrasing, hallucination safeguards (LLM restricted to restating structured evidence only).
- **Figure 2**: Detailed architecture diagram (centerpiece figure).
- **Figure 3**: End-to-end sequence diagram for one sample query (e.g., combined diabetes+heart query), showing data transformation at each stage.
- **Table 2**: Module summary — Module ID, Disease, Model Type, Input Modality, Dataset, Output.
- **Table 2b**: Module | XAI Methods Used | Output Type | Fed to LLM as (structured block).

### 4. Datasets (~0.75-1 page)

- Source, size, features/specs, class distribution, split — per disease; hospital collaboration status.
- **Table 3**: Dataset summary (Dataset Name, Source, #Samples, Modality, Classes, Split Ratio).
- **Figure 4 (optional)**: class distribution bar chart, if imbalance is notable.

### 5. Experimental Setup (~0.5 page)

- Hardware, frameworks, hyperparameters.
- Metrics: accuracy, precision, recall, F1, AUC-ROC, Expected Calibration Error (ECE) for uncertainty validation.

- Report evaluation protocol: human raters, rubric (factual consistency, clarity, hallucination rate, usefulness).

## 6. Results and Discussion (~2-2.5 pages)

- 6.1 Per-module ML performance.
  - **Table 4:** all 9 modules — Accuracy, Precision, Recall, F1, AUC.
  - **Figure 5:** grouped bar chart, accuracy/F1/AUC across modules.
  - **Figure 6:** ROC curves or confusion matrix grid.
- 6.2 Explainability validation.
  - **Figure 7:** SHAP summary plot (1-2 diseases).
  - **Figure 8:** SHAP-vs-LIME agreement plot (rank correlation of top features) — cheap, genuinely novel result.
  - **Figure 9:** Grad-CAM / Integrated Gradients heatmap for an image module.
- 6.3 Uncertainty and calibration.
  - **Table 5:** ECE / calibration results per module, before vs. after calibration.
- 6.4 LLM report quality.
  - **Table 6:** Report scores — Factual Consistency, Clarity, Hallucination Rate, Clinician Usefulness (mean  $\pm$  SD).
  - **Figure 10:** radar chart of report quality dimensions, or hallucination rate bar chart by disease.
- 6.5 Confidence-aware behavior.
  - **Table 7:** examples of high- vs. low-confidence cases and how report language changed — strong novelty demonstrator, keep even if small.
- 6.6 Sample end-to-end outputs — 2-3 full example reports (single-disease + multi-disease combined), shown as boxed text/figure.
- 6.7 Discussion — what worked, failure cases, where routing/confidence logic broke.

## 7. Limitations and Ethical Considerations (~0.4-0.5 page)

Public dataset dependence, no clinical trial validation, residual hallucination risk despite safeguards, demographic bias if known, data privacy/consent for hospital collaboration, explicit "decision-support not diagnostic replacement" framing.

## 8. Conclusion and Future Work (~0.3-0.4 page)

Summary, planned clinician-in-the-loop evaluation, real hospital data validation, multilingual/voice expansion, federated learning across hospitals for privacy-preserving training.

References (25-35 entries)

Full table/figure checklist

| #         | Item                                                           | Section |
|-----------|----------------------------------------------------------------|---------|
| Table 1   | Comparative summary of related systems                         | 2       |
| Table 2   | Module summary (ID, disease, model, modality, dataset, output) | 3       |
| Table 2b  | XAI methods per module → LLM input format                      | 3.4.3   |
| Table 3   | Dataset summary                                                | 4       |
| Table 4   | Per-module ML performance metrics                              | 6.1     |
| Table 5   | Calibration/ECE results                                        | 6.3     |
| Table 6   | LLM report quality scores                                      | 6.4     |
| Table 7   | High- vs. low-confidence report language examples              | 6.5     |
| Figure 1  | System teaser diagram                                          | 1       |
| Figure 2  | Full architecture diagram                                      | 3       |
| Figure 3  | End-to-end query sequence diagram                              | 3       |
| Figure 4  | Class distribution (optional)                                  | 4       |
| Figure 5  | Accuracy/F1/AUC bar chart, all modules                         | 6.1     |
| Figure 6  | ROC curves / confusion matrices                                | 6.1     |
| Figure 7  | SHAP summary plot                                              | 6.2     |
| Figure 8  | SHAP-vs-LIME agreement plot                                    | 6.2     |
| Figure 9  | Grad-CAM/Integrated Gradients heatmap                          | 6.2     |
| Figure 10 | Report quality radar/bar chart                                 | 6.4     |

### What makes this genuinely novel — completion checklist

| Must-have                                                                | Why it matters                                                            |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Real confidence-weighted routing between modules, not just parallel runs | Most multi-disease papers fake orchestration                              |
| Multi-method XAI fused into one structured evidence block per module     | SHAP-only or LIME-only is now common; fusion + agreement analysis is rare |

| Must-have                                                                    | Why it matters                                            |
|------------------------------------------------------------------------------|-----------------------------------------------------------|
| Counterfactual explanations feeding the LLM                                  | Gives actionable, not just descriptive, output            |
| Guideline-grounded phrasing (retrieval from fixed clinical guideline corpus) | Strong hallucination safeguard, genuine novelty           |
| Measured hallucination rate                                                  | Almost never quantified in comparable papers              |
| Cross-module correlation reasoning (e.g., diabetes+kidney link)              | Proves the "agent" label is functional, not just branding |
| Human evaluation of report quality (even 5-10 raters, 40-50 reports)         | Turns "we built it" into "we validated it"                |
| Calibration analysis (ECE) instead of raw softmax confidence                 | Preempts the most common reviewer criticism in medical ML |

Given limited time, prioritize in this order: hallucination rate → SHAP-vs-LIME agreement → confidence-to-language examples → guideline grounding → counterfactuals. The first three are cheap and are exactly what reviewers say is missing from similar work.

## Completely AI generate (Not humanized)

### Abstract (draft)

Modern healthcare AI systems largely operate as isolated, single-disease classifiers that output raw probabilities without clinically interpretable context, leaving patients and clinicians to interpret results with little guidance. This paper presents an agentic, multi-modal framework that integrates nine disease-specific machine learning modules—spanning diabetes, heart disease, pneumonia, brain tumor, breast cancer, liver disease, kidney disease, skin cancer, and eye disease—into a unified diagnostic pipeline orchestrated by a large language model (LLM) acting as a coordinating agent. The system accepts multi-modal patient input (text, images, and audio), routes it to relevant disease modules, and enriches each module's prediction with a fused explainability layer combining SHAP, LIME, counterfactual explanations, feature interaction analysis, and calibrated uncertainty estimates. This structured evidence is passed to the LLM agent, which synthesizes cross-module findings, grounds its language in clinical guideline references, and generates a final, confidence-aware report for the patient or physician. Unlike prior systems that apply a single explainability method or treat each disease module independently, our framework introduces evidence fusion across multiple explanation techniques and explicit cross-module correlation reasoning (e.g., linking diabetic and renal risk signals). We evaluate the nine modules on [accuracy/F1/AUC ranges], and assess LLM-generated reports on factual consistency, clarity, and hallucination rate through human evaluation. Results show [placeholder for key finding], demonstrating that agentic orchestration with multi-method explainability can produce medical reports that are both accurate and interpretable. The proposed system is intended as a clinical decision-support tool to assist—not replace—medical professionals, with implications for scalable, explainable AI deployment in resource-constrained healthcare settings.

*(Fill in the bracketed results once you have final numbers — keep this to ~200-250 words in the final version; the draft above is slightly long so you can trim to what fits your template.)*

### 1. Introduction (draft)

Artificial intelligence has demonstrated strong diagnostic performance across a wide range of medical conditions, from diabetic risk prediction using tabular clinical data to tumor detection from radiological images. However, most existing systems are narrow in scope: they are trained and evaluated on a single disease, accept only one input modality, and return a bare probability score with little explanation of the reasoning behind it. In real clinical workflows, patients rarely present with a single, cleanly defined concern, and clinicians require not just a prediction but a transparent account of the evidence supporting it. This gap between fragmented, single-purpose AI tools and the multi-condition, evidence-based reasoning that clinical practice demands motivates the framework proposed in this paper.

We introduce a system that unifies nine disease-specific prediction modules—covering metabolic, cardiac, oncological, hepatic, renal, dermatological, ophthalmic, and radiological conditions—under a single orchestrating layer powered by a large language model acting as an autonomous agent. Rather than simply summarizing one model's output, the LLM agent receives structured, multi-method explainability evidence from each triggered module, reasons across modules when clinically relevant signals co-occur, and produces a final report whose language reflects the calibrated confidence of the underlying predictions.

The contributions of this work are as follows:

1. **Multi-modal input ingestion**, allowing patient or clinician queries to be expressed as free text, spoken audio, or medical images, which are parsed and routed to the appropriate disease modules.
2. **A nine-module diagnostic backbone** covering diabetes, heart disease, pneumonia, brain tumor, breast cancer, liver disease, kidney disease, skin cancer, and eye disease, each trained and validated independently.
3. **A fused, multi-method explainability layer** that combines SHAP, LIME, counterfactual explanations, feature interaction analysis, and similar-case retrieval into a single structured evidence representation per module, rather than relying on one explanation technique in isolation.
4. **An LLM-as-agent orchestration mechanism** that performs confidence-aware routing, cross-module correlation detection (e.g., recognizing shared risk between diabetes and kidney disease), and guideline-grounded report synthesis.
5. **A hallucination-aware, calibrated reporting pipeline**, in which the LLM is constrained to reason only over supplied structured evidence, and where predictive confidence is estimated using calibration techniques rather than raw softmax scores.
6. **An end-to-end evaluation methodology** that assesses not only per-module predictive performance but also the factual consistency, clarity, and hallucination rate of the generated clinical reports through human evaluation.

The remainder of this paper is organized as follows. Section 2 reviews related work in disease-specific machine learning, explainable AI, and LLM-based medical agents.

Section 3 details the proposed system architecture.

Section 4 describes the datasets used.

Section 5 outlines the experimental setup.

Section 6 presents results and discussion.

Section 7 addresses limitations and ethical considerations, and

Section 8 concludes with directions for future work.